

Trainings ▾

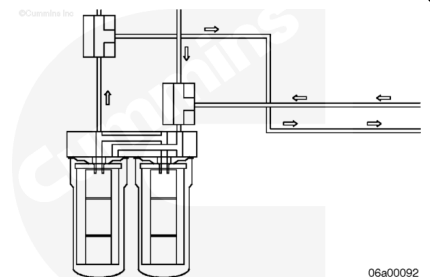
General Information

With Electric Lift Pump(s)

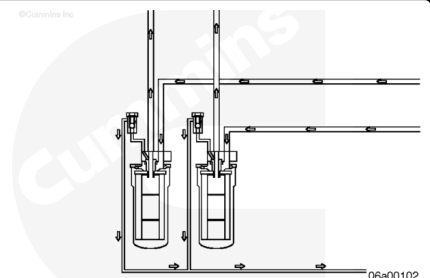
There are two variations of the electric fuel lift pump system – one utilizing two electric fuel lift pumps and the other with three electric fuel lift pumps. This procedure applies to both variations.

With Mechanical Lift Pump(s)

The QST30 uses two different types of fuel filter heads. The first type of fuel filter head is a two-filter head with one inlet and one outlet. In this configuration, the fuel is “shared” by the two filters. This means that the left-bank and right-bank fuel is mixed at the fuel filters.

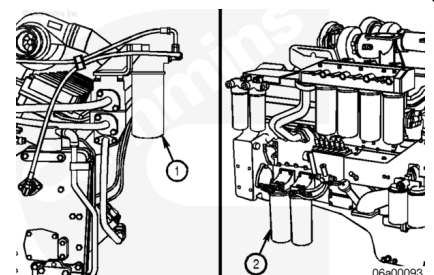


The second type of fuel filter head is a two-filter split-system fuel filter head. In this system, there are two inlets and two outlets, one for each filter. In this configuration, the fuel is not “shared,” or mixed, at the fuel filters.

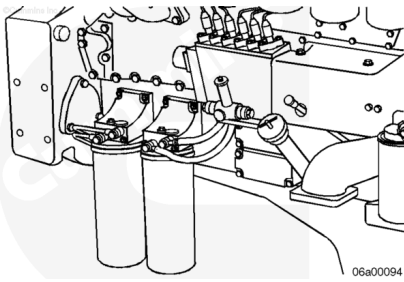


The industrial engine can have either type of fuel filter head configuration positioned at the following locations:

1. Type 1 (1) is located at the left-bank aftercooler.
2. Type 2 (2) is located at the right-bank rear cylinder block.



The generator set/generator-drive engine has only the type 2 fuel filter head. It is positioned on the right bank rear of the cylinder block.



Remove

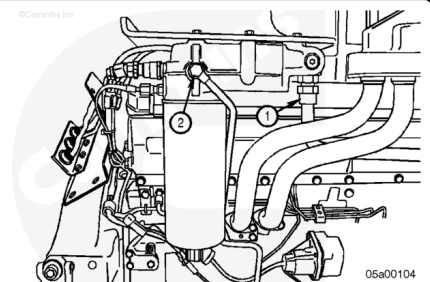
With Electric Lift Pump(s)

Prior to fuel filter removal for maintenance interval, prime the fuel system. Refer to Procedure 005-234 in Section 5. (</qs3/pubsys2/xml/en/procedures/57/57-005-234.html>)

Remove the fuel filter.

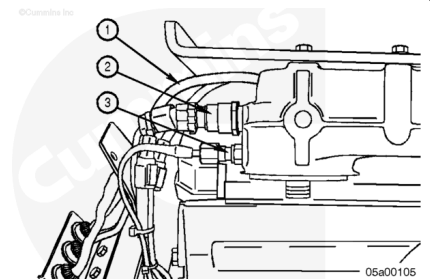
Disconnect the fuel lines.

After removing each component, cover all open ports and connections using protective caps from Fuel System Clean Care Kit, Part Number 5394230.

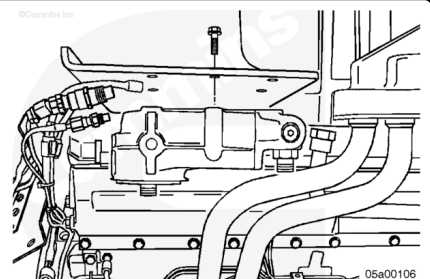


Disconnect the fuel lift pump engine harness connectors (1).

Disconnect the fuel pressure (2) and fuel temperature (3) sensor engine harness connectors, if equipped.



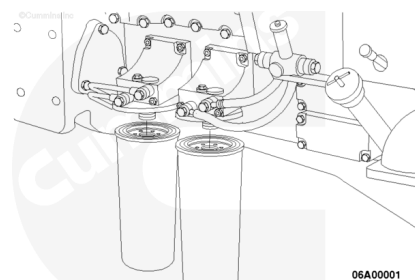
Remove the lift pump housing from the mounting bracket.



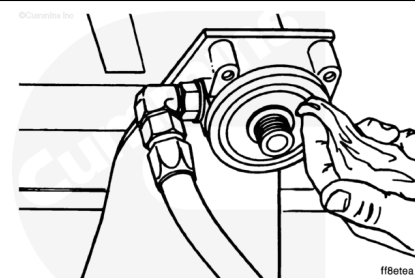
With Mechanical Lift Pump(s)

Clean the area around the fuel filter head and filter.

Remove the fuel filter with filter wrench, Part Number 3375049.

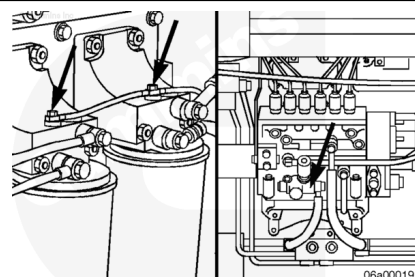


Use a clean, lint-free towel to clean the filter head gasket surface.



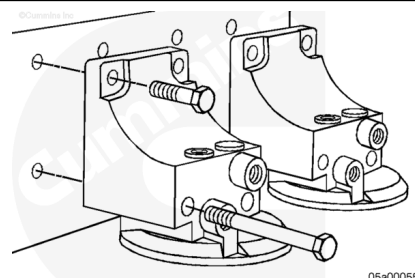
Remove the fuel vent line and inlet/outlet lines from the fuel filter head (used on generator-drive/generator set, **only**).

After removing each component, cover all open ports and connections using protective caps from Fuel System Clean Care Kit, Part Number 5394230.

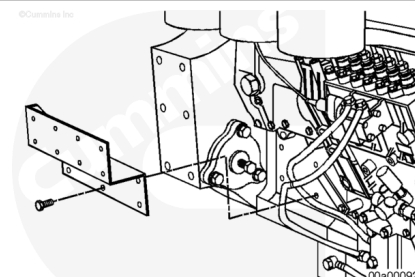


Remove the four mounting capscrews.

Remove the fuel filter head from the bracket.



Remove the capscrews and the fuel filter head bracket.



Clean and Inspect for Reuse

With Electric Lift Pump(s)

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a cleaning solvent that will not harm aluminum.

Use solvent to clean the filter head assembly.

Dry with compressed air.

Inspect the filter head for cracks and damaged threads.

Check the wiring harnesses for cracks, wear, or damaged pins.

With Mechanical Lift Pump(s)

⚠ WARNING ⚠

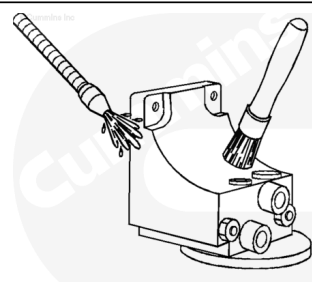
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a cleaning solvent that will not harm aluminum.



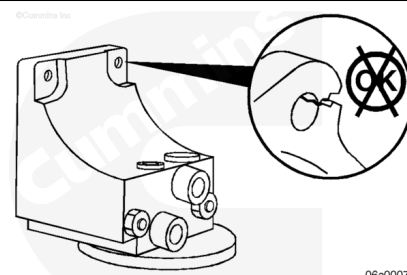
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Clean the fuel filter head with solvent.

Use compressed air to dry the fuel filter head.

Inspect the fuel filter heads for cracks, damaged mounting bolt holes, and any other damage.

Note : If any cracks are found, the fuel filter heads must be replaced.

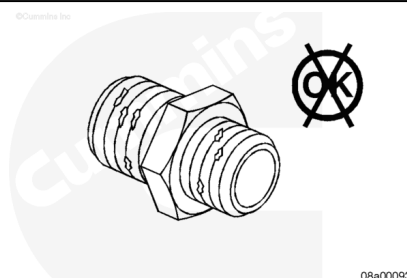


Inspect the threads of the filter head inserts for damage; replace if necessary.

Apply Loctite™ 277 or equivalent to the threads in the filter head.

Install the filter head insert.

Torque Value: 112 n•m [83 ft-lb]

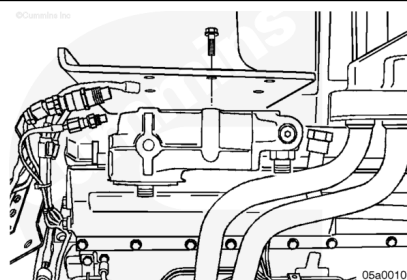


Install

With Electric Lift Pump(s)

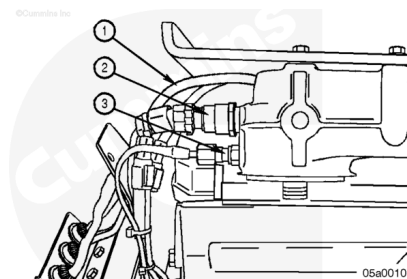
Mount the lift pump housing to the mounting bracket with five capscrews.

Torque Value: 67 n•m [49 ft-lb]



Connect the fuel pressure (2) and fuel temperature (3) sensor engine harness connectors, if equipped.

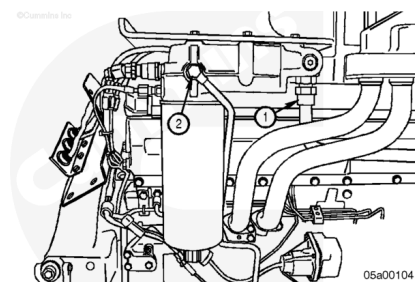
Connect the fuel lift pump engine harness connectors (1).



Connect the fuel lines (1, 2).

Torque Value: 40 n•m [30 ft-lb]

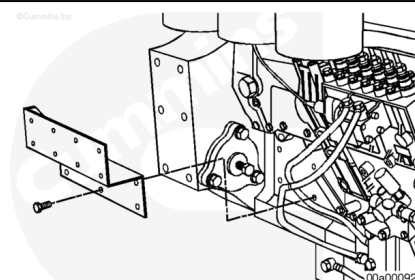
Install the fuel filter.



With Mechanical Lift Pump(s)

Install the fuel filter head bracket and capscrews.

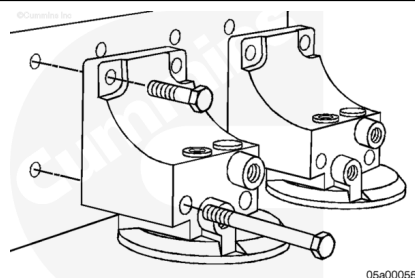
Torque Value: 65 n•m [48 ft-lb]



Install the fuel filter head onto the bracket.

Install the four mounting capscrews.

Torque Value: 65 n•m [48 ft-lb]



Install the fuel vent line from the fuel filter head (used on Generator-drive, Generator-set, **only**).

Check that the sealing washers are on both sides of the banjo connector capscrews.

Torque Value: 8 n•m [71 in-lb]

After fuel lift pump is installed and fuel system is connected, prime the fuel system. Refer to Procedure 005-234 in Section 5. (</qs3/pubsys2/xml/en/procedures/57/57-005-234.html>)

